Applied Flood Modelling System For Flood Inundation Simulation

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Executive Summary

The intention behind the Applied Flood Modelling System for Flood Inundation Simulation project was to address the need that communities face in assessing their flood risk. Namely, the conventional offerings are incredibly expensive and take a significant amount of time to conduct a flood risk assessment. In the face of growing threats due to flooding under climate change, we sought to build a Flood Information System (FIS) that can be simple, cheap, and adaptive to the needs of individual communities so they can make informed decisions to mitigate their flood risk.

The main goals of the project are as follow:

- Exploring simplified flood models and their ability to assess flood risk
- Incorporate low-cost and consumer hardware, such as the iPad Pro and Livox Mid-40, into the FIS. This will allow users of the FIS to collect high quality data at a lower cost, as well as incrementally add to the total dataset.
- Investigate the use of emerging interactive technologies, such as Virtual Reality, for providing knowledge and information of floods to public and professional users.

Overall, the project was successful in initiating the development of an integrated Above Atlantis Flood Information System. Over the short project duration, a comprehensive assessment of existing FIS was conducted. Additionally, a prototype of a LiDAR sensor package was developed that can be built for under $1,500 and can scan objects up to 200 metres away. Building of the floodplain in the Unity 3D game engine has commenced.

Key benefits to Above Atlantis include increased understanding of the needs of flood-stricken communities, connections to academic experts in the field, and access to technical equipment, such as VR headsets and LiDAR scanners. The Mitacs eAccelerate grant enabled the student founders of Above Atlantis to have the valuable opportunity to have their work be funded.

Background Information

Flood risk characterization and visual communication are critical to develop understanding and action plans for communities across Canada. According to the Association of Professional Engineers and Geoscientists of BC, only 21% of communities affected by floods have up-to-date flood maps (APEGBC, 2017). This highlights the need to develop an affordable, accessible and effective flood modelling technique for the British Columbian context.

Effective flood modelling requires high fidelity data of the geometry of the floodplain. Light Detection And Ranging (LiDAR) technologies have been widely adopted in surveying of flood plains (Wedajo, 2017). Mobile Laser Scanning (MLS) is an advancement of LiDAR technology that has been able, when coupled with hydro-acoustic techniques, to dynamically monitor changes in floodplain landscape due to erosion (Leyland et al., 2017). This work demonstrates that LiDAR can produce large quantities of high-fidelity data, at high temporal and spatial resolution. When mounted on a range of platforms, including aerial drones, LiDAR allows incredibly efficient and effective collection of spatial data.

The costs of both Mobile Laser Scanning (MLS) and Terrestrial Laser Scanning (TLS) have steadily decreased in recent years, increasing their accessibility. However, conventional LiDAR technology is still out of reach for many individuals and organizations. There is a need to investigate the viability of lower-cost and consumer-grade LiDAR technology, such as those equipped on the iPhone 12 Pro and iPad Pro. In addition, LiDAR sensors developed primarily for self-driving vehicle applications have become significantly less expensive, and may be able to be re-utilized for capturing 3D spatial data and surveying activities.

There is a research need to develop new flood modelling frameworks. Currently, one-dimensional, two-dimensional and three-dimensional flood modelling techniques are used predominantly by environmental consulting agencies (Brunner, 2016). These models provide highly accurate flood maps when quality data is available. However, these flood models require large amounts of input data, which may not be available in many cases (Savage et al., 2016). In addition, these complex models can be very computationally expensive, limiting their use in near-real time models (Savage et al., 2016).

To address this need, newer models, like the Height Above Nearest Drainage (HAND) model, are being developed to provide accurate results with fewer data inputs (McGrath et al., 2018). The HAND model uses primarily the Digital Terrain Model (DTM) of a study area as its input (A. D. Nobre et al., 2011). By using only the DTM data sets, the HAND model can be used in study areas that have limited data available, such as soil types (A. D. Nobre et al., 2011). The DTM data is readily collected by handheld or aerial platforms, like drones (Wedajo, 2017). The HAND model has been demonstrated to be effective in various watersheds throughout the world, including Utah, USA (Garousi-Nejad et al., 2019), Gatineau, QC and Fredericton, NB (McGrath et al., 2018), and Santa Catarina, Brazil (Antonio Donato Nobre et al., 2016). The HAND model has yet to be applied to the context of communities in British Columbia however, indicating a research need.

Three-dimensional visualizations of the effects of floods on communities have been shown to be effective in motivating decision making around flood mitigation efforts (Adda et al., 2010). Traditional paper maps are not easily accessible to laymen audiences, and do not provide dynamic and interactive experiences that can provide increased understanding of the flood phenomena (Adda et al., 2010). Providing these highly effective visualizations and data can have a wide range of uses, including the provision of risk reduction tools to disaster managers during flood events (Zerger & Wealands, 2004). Three-dimensional visualizations have been shown to be effective at communicating risk due to sea-level rise and erosion in the province of Prince Edward Island (Fenech et al., 2017). In particular, the CoastaL Impacts Visualization Environment (CLIVE) developed by Fenech et al. prioritized the integration of the powerful analysis of coastal erosion with photos and visualizations that allowed for users to connect intuitively and emotionally with the results. Interactive, dynamic, analytical and intuitive use of geovisualizations empower users to explore the results themselves, gleaming new understanding of the phenomena.

This research project will explore lower-cost LiDAR technology alternatives to equip more individuals and organizations with the ability to conduct quality spatial data and geometry capture. This work will then expand on the effectiveness demonstrated by the HAND model in existing literature. Highly accurate maps have been produced by the outputs of HAND models, but we would like to develop more automated methodologies that can further increase the speed and effectiveness of the implementation of these flood models for mapping purposes. Finally, the project will use advanced visualizations to make the information accessible to the communities impacted by flooding.

STUDY GOALS

The general objective of the research project is to explore the multi-faceted process around modelling and visualizing flood data. Ultimately, the goal of the research is to multiply the utility of existing data sets, in the form of LiDAR point clouds provided by the Fraser Basin Council and the Okanagan Basin Water Board. The goals of the research can be organized into two areas: the characterization of phenomena, and the visualization and interpretation of data and models.

The characterization of phenomena refers to an entire process, including the collection of data in the field, data cleaning and filtering, and integrating the spatiality of the data. The initial goals as outlined in the proposal included the following:

- Utilizing the Height Above Nearest Drainage (HAND) model and exploring its flexibility as a simplified model.
- Exploring the factors that affect the output of the HAND model, through the development of multiple permutations of the model.
- Acquire knowledge on the integration of machine learning with flood models, to take the inundation extent of historic flood events and inform the model's parameters.

This project served an exploratory role in using simplified flood models. While we were able to conduct a basic flood risk analysis using the HAND model in the Okanagan Valley, more work would need to be done to confirm its veracity compared to more conventional flood models.

To study the visualization and interpretation of flood data and models, we laid out the following goals:

- Investigate how to build visualizations that allow users to explore different HAND model outputs and conduct comparative analysis between the models.
- Understand how to create visualizations that can facilitate understanding of the flood models to a range of audiences, including laypersons and professionals in the field.
- Explore the usefulness of visualizations in informing working professionals in flood mitigation and response, including first responders.

The goal of our literature review is to understand the landscape of flood decision support systems (FDSS) and integrated flood information systems (IFIS). There are some characteristics that these systems share, including their ability to provide a high-level understanding of the behaviour of floods and their impact on the area of interest. There is variation in the features and abilities that these systems have, ultimately affecting the utility they can provide for users. The intended users can vary as well, from simple informational resources for residents in flood-prone areas to engineers planning flood mitigation infrastructure.

Our review is intended to capture the best available understanding for how to build a flood information system to inform the design of our own prototype. The review was successful in surveying and capturing the current practices and trends in academic and applied flood information systems. This project identified the gaps and needs in practice and capability in flood risk visualization that may represent research need and market opportunity.

Literature review

We conducted a systematic literature review using the Web of Science database, and ultimately analyzed 67 different flood information systems. Flood risk management is empowered by information systems at different levels. We found that there are primarily three phases where information systems and modelling tools are used for flood risk management. The first is as decision making tools. These are primarily qualitative models that work with resilience and risk to prioritize areas for future action. These models or frameworks primarily concern themselves with the organization of thinking and people involved in the decision making process around floods. The second kind of information systems are in the form of real or near-real time hydrological monitoring. These systems serve to provide data and information as quickly as possible to assist in immediate response to a flood event. The final kind of information systems exist to analyze the past in detail and support informed decision making. They are more quantitatively detailed than the other two and are often conducted with data collected in retrospect. The aim of these systems is to better understand past flood events and predict future scenarios. The infrastructure and analytical demand becomes progressively more for the information system, in the order discussed here.

The amount of data required to fuel an information system, depends on the type or use-case for the information system. The use of available real or near-real time data can be used very quickly and almost instantaneously (Chang et al., 2012; Heil et al., 2014; Mandl et al. 2013; Singh et al., 2017; Waser et al. 2018). Forty-three studies we reviewed included support for real-time data integration. Thirty-seven of the studies included river flow data, 12 included snowpack height, and 38 included the estimated extent of flood inundation in given scenarios.

The literature has a strong focus on the use of web-based GIS interfaces for the delivery of spatial information. This is often paired with other data output types as well, including graphs and tables. Karavokiros et al. (2016) and Fertier et al. (2020) discuss the architecture of such web-accessible information structures. Fertier et al. (2020) and Balis et al. (2018) focus on evaluating decisions in the deployment of such information systems for efficient performance. Information systems benefit from being web-accessible as they are able to quickly provide up-to-date information to users across the field. This is also useful when the system is brought in for discussion in decision making processes. Depending on the audience, the amount of information and interactivity vary.

Using our the criteria and rubric developed for our methodology, we found that 14 studies have low level of interactivity, 7 studies have low-medium level of interactivity,14 studies have medium level of interactivity,8 studies have a medium-high level of interactivity, and 10 studies have a high level of interactivity. For methods of presentation of information, we found that 37 included graphs, 45 included

2D maps, 8 included 3D visualizations, 37 included written descriptions, and 29 included tables in their flood information systems. There is a heavy focus on the use of 2D mapping for the communication of spatial data, with data presented in graphs and tables serving as a supplement for use by trained technicians.

We found little information of the applicability of created systems, with very limited usability feedback. Despite the purpose of a flood information system, communication did not seem to be the main focus. Current flood information systems do not prioritize the accessibility of knowledge and understanding, with most systems merely providing data to users that already have the relevant knowledge context to effectively work with what is provided by the FIS. Most of the FIS interfaces are built with WebGIS, which can pose a barrier for people without previous GIS training.

LiDAR sensor package

At the center of the hardware package is the LiDAR sensor, which is a Livox Mid-40 unit. This is a relatively low-cost high-fidelity LiDAR sensor manufactured originally for use on autonomous vehicles. Significant investment in autonomous vehicles have led to the reduction in costs for LiDAR sensors for the purpose, compared to more specialized LiDAR devices designed for surveying use.

The Raspberry Pi 4 is the main computational unit in the hardware package, and its main purpose is to relay the sensor outputs and connect to other computers. This allows the LiDAR sensor package to interface with other computers on its network wirelessly, enabling remote functionality. The SparkFun Auto pHAT was selected as the inertial measurement unit, for its plug-and-play support of the Raspberry Pi. Additionally, it gives the Pi the ability to control electrical motors, which could allow automated scanning in future versions. GPS capabilities were added to the hardware package via the Adafruit Ultimate GPS FeatherWing, with the SparkFun ESP32 Thing.

To control and manage these sensors, the Robot Operating System (ROS) was selected. ROS is an open-source middleware used in robotics, allowing hardware built by different manufacturers to communicate and for the developers to control their robots. Additionally, ROS has significant community and industry support. ROS is also capable of interfacing with the Unity 3D game engine, for visualization purposes in the Above Atlantis FIS.

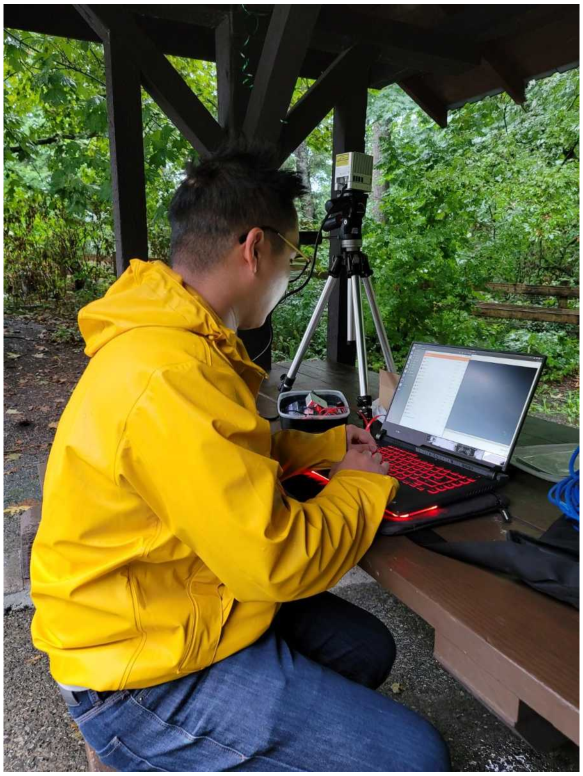

Figure 1. This image captures the student intern setting up the low-cost LiDAR unit, interfacing with a laptop.

Figure 2. This image shows the low-cost LiDAR unit set up in the field, scanning a culvert. This serves as an example of the low-cost LiDAR abilities to capture small-scale flood mitigation for integration into the flood information system.

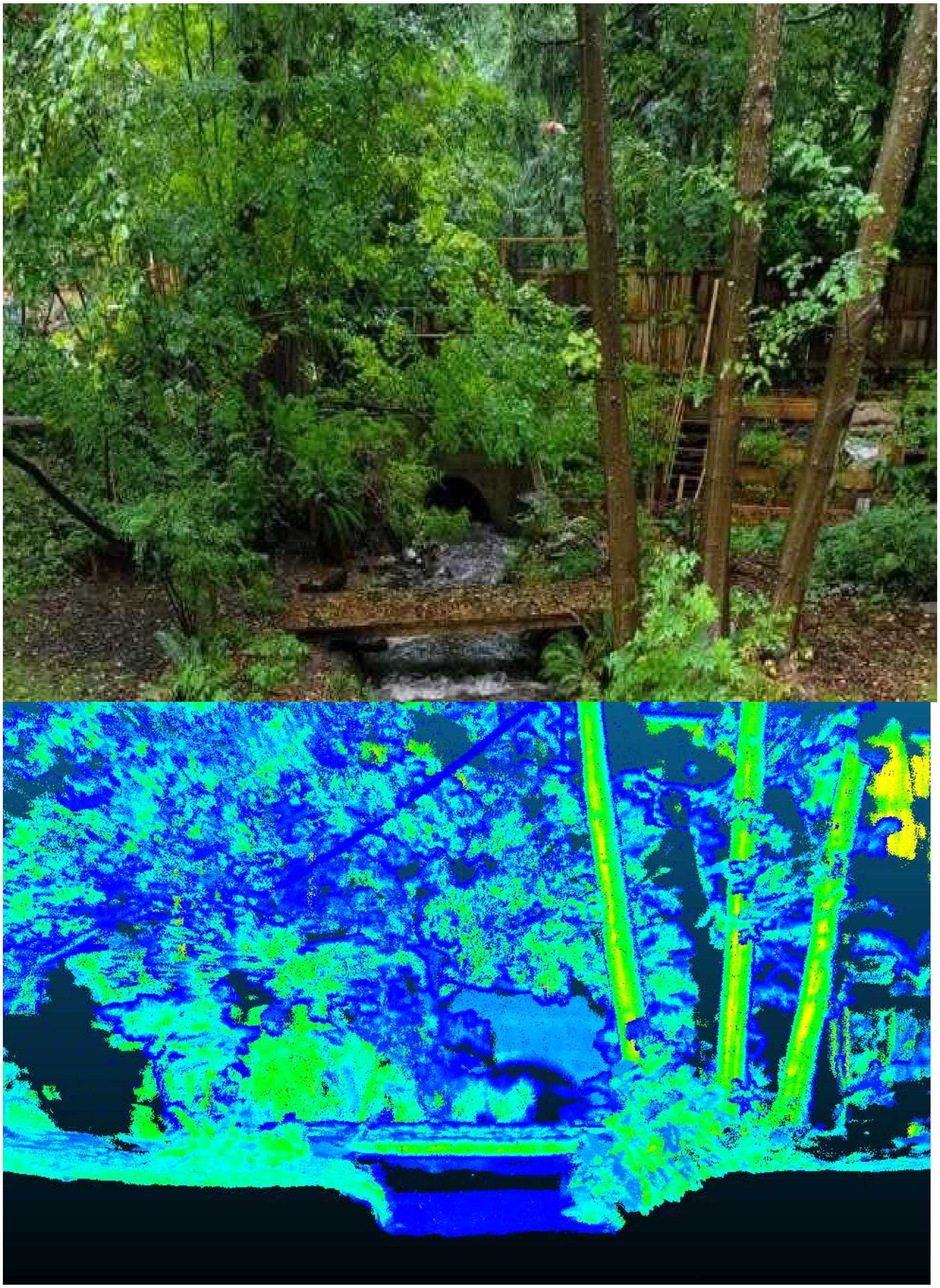

Figure 3. Comparison of standard visual light camera and screenshot of LiDAR-derived three-dimensional point cloud. The bridge and tree features are particularly distinct in the LiDAR scan, but also the concrete culvert in the background can be resolved.

LiDAR

We developed a low-cost LiDAR sensor package in order to increase the accessibility of data collection tools to enable users of the Above Atlantis FIS to contribute their own data. The LiDAR sensor package was assembled from technology that would be available at low-cost and be agile in its deployment, not requiring specialized skills or equipment.

The main priorities for the development of the was accessibility and agility. By relying on open-source software, we could ensure reproducibility, maintainability, and accessibility for other researchers and developers. Using off-the shelf components for hardware allowed our work to be easily replicated and scaled.

Table 1. *Description of information / variables collected from flood information systems*

Category	**Variable**	**Description**
Basic information	Name of the system	Name of the flood information system
	Organization / Person that developed the system	Organization or person that developed the system May include the author of the paper, in addition to the organization that they worked with to produce the system (ex. Government agency, university, municipality)
	Original location for which the system was developed	Geographic location that the flood information system covers. This may contain information about the size of the area covered by the system as well.
	Brief summary / description of purpose of system	Description or summary of the purpose of the system or the motivation behind the development of the system. May include the context with current gaps in flood information accessibility, which the FIS would attempt to address.
	Target audience	Description of the typical or intended user of the FIS. May include some description of the users' needs that the FIS attempts to address.

Information types included in the system	Brief description of hydrological context (type of flooding, rural/urban, ideal watershed/channel characteristics)	Description of the hydrology that the FIS attempts to capture. May include some description of what features that the FIS is limited to.
	River flow data (Y/N)	Yes / No answer of whether the FIS shows river flow data to the user. The source of this data could be from hydrometric stations, which often provide near-real time data.
	Snowpack height (Y/N)	Yes / No answer of whether the FIS shows snowpack height data to the user.
	Flood inundation extents based on probability (Y/N)	Yes / No answer of whether the FIS shows flood inundation extents based on scenarios associated with probabilities, usually described as Annual Exceedance Probabilities (AEP).
	Other information:	Include other information that is provided to the user through the FIS.
Information presentation methods	Graphs (Y/N)	Yes / No answer to whether the FIS uses graphs to display information to the user. Include description of what information is displayed to the user through graphs.
	2D Maps (Y/N)	Yes / No answer to whether the FIS displays information on 2D maps, usually in the form of layers. May include addition description of whether using vector or raster data formats. Include description of what information is displayed to the user through 2D maps.
	3D Visualizations (Y/N)	Yes / No answer to whether the FIS displays information using 3D visualizations. Include description of what information is displayed to the user through 3D visualizations.
	Written descriptions (Y/N)	Yes / No answer to whether the FIS provides information to the user through written descriptions. Include description of what information is provided to the user through this method.
	Tables (Y/N)	Yes / No answer to whether the FIS displays information in tables. Include description of what information is provided through tables.
	Other	Include description of other methods of data presentation.

Situatedness	In situ or ex situ?	Situatedness describes whether the user's location is an important part of the FIS function. In situ is where the user's position is described in the FIS. For example, a FIS that queries and changes the information presented based on the user's location would be considered an in situ FIS. Ex situ is where the user's position is not part of the use of the FIS.
	Description of situatedness of the information system	Describe further how the FIS is either in situ or ex situ, or which features of the FIS fall under each.
Interactivity	Level of interactivity (1 to 5)	Describes roughly the interactivity of the FIS. Score of 1 corresponds to static presentation of information, where the user has no ability to change the presentation of the information based on their needs. For example, these would be static PDF maps. Score of 5 corresponds to dynamic presentation of information, where the user has the ability to change the presentation of the information, leading to knowledge creation. For example, these could be VR experiences where the user can explore the geovisualization.
	Collaborative tools? (Y/N)	Collaborative tools allow more than one user to access the FIS at the same time to contribute to one presentation of the information.
	Describe interactivity	Describe further what the features of interactivity and justification for the interactivity score.
Structure / framework	Does the study describe how the system is set up? (cloud services, coding languages, data structures, etc.) (Y/N)	Yes / No answer of whether the study describes how the system is set up and how different components of the system interact and connect.
	Short description of system structure	Describe the system components and how these are structured to ultimately provide the relevant flood information to the user.
	Screenshot of chart showing how system was built	Provide a screenshot of a chart, if available, from the paper that describes how the FIS is set up.

Reviewing the collected data, we were able to analyze trends in the FIS space. We were able to compare the strengths and weaknesses, as well as the lessons learned from the creators of each FIS. Using these notes, we can direct the development of the flood information system for Above Atlantis.

Flood Information System - Data Presentation

For the development of the Flood Information System, we identified a list of features and quests that we would like to accomplish in the prototype. This list was generated based on the findings from the literature review, as well as the desire to explore the use of emerging interactive technologies for flood information presentation.

An important aspect to our approach with the development of the FIS is to ensure it is designed user-first rather than information first. We must consider what the users of the FIS actually need to know, rather than providing all the available information at once and placing the cognitive load on the user to correctly contextualize the information.

Virtual Reality experiences provide immersive and engaging visualizations that can convey more information in terms of quantity, as well as more intuitive understanding of information through the 3D interactive interface. The 3D interface can provide multiple layers of information that can be toggled and manipulated for knowledge creation. The user should be able to access a navigable environment of the watershed. The FIS should be able to be used in situ, so that it can be used on mobile devices in the field to verify real world features with what is contained in the flood information system.

In terms of data collection features, we would like to have the ability to add data collected by users into the digital twin model in the Unity 3D space. We will continue to investigate the use of handheld devices for simplified laser scanning with consumer-grade hardware, like the iPad Pro, to generate .obj that could be then imported into the larger world.